

Group6_Part2

Authors: Hilde Gelinck, Niels Pleijsier and Jesse de Heus

Load data

```
library(tidyverse)
library(rsample)
library(caret)
library(rms)

# Read fresh
smart1 <- read.csv2("smartc.csv")

# Convert the true categorical variables to factors FIRST
smart1 <- smart1 %>%
  mutate(
    EVENT      = factor(EVENT,      levels = c(0,1),  labels = c("no","yes")),
    DIABETES    = factor(DIABETES,  levels = c(0,1),  labels = c("no","yes")),
    CEREBRAL    = factor(CEREBRAL,  levels = c(0,1),  labels = c("no","yes")),
    CARDIAC     = factor(CARDIAC,   levels = c(0,1),  labels = c("no","yes")),
    AAA         = factor(AAA,       levels = c(0,1),  labels = c("no","yes")),
    PERIPH      = factor(PERIPH,    levels = c(0,1),  labels = c("no","yes")),
    STENOSIS    = factor(STENOSIS,  levels = c(0,1),  labels = c("no","yes")),
    SEX         = factor(SEX,       levels = c(1,2),  labels = c("male","female")),
    SMOKING     = factor(SMOKING,   levels = c(1,2,3), labels = c("never","former","current")),
    alcohol     = factor(alcohol,   levels = c(1,2,3), labels = c("never","former","current")),
    albumin     = factor(albumin,   levels = c(1,2,3), labels = c("no","micro","macro"))
  ) %>%
  # Decimal columns read as text -> numeric. The factors above are left untouched.
  mutate(across(where(is.character), as.numeric))
```

```
# Sanity checks
stopifnot(is.numeric(smart1$BMI), sum(is.na(smart1$EVENT)) == 0)
str(smart1[c("EVENT", "BMI", "HDL", "IMT", "albumin")]) # EVENT/albumin factor; BMI/HDL/IMT

'data.frame':  3873 obs. of  5 variables:
 $ EVENT   : Factor w/ 2 levels "no","yes": 1 1 1 2 1 1 2 2 2 2 ...
 $ BMI     : num  23.9 24 26.2 29.6 29.6 ...
 $ HDL     : num  0.94 1.26 1.28 1 0.81 0.95 0.87 1.32 0.58 0.95 ...
 $ IMT     : num  0.82 0.57 0.83 1.45 1.07 0.97 1.15 0.7 1.48 0.87 ...
 $ albumin : Factor w/ 3 levels "no","micro","macro": 1 1 1 2 1 3 1 2 1 1 ...
```

Logistic Regression Model

```
model <- glm(EVENT ~ AGE + SEX + BMI + SYSTH + HDL + DIABETES + HISTCAR2 +
             HOMOC + log(CREAT) + STENOSIS + IMT + SMOKING + alcohol + albumin,
             data = smart1, family = binomial)
```

Question A

The performance of the logistic regression model is evaluated using 5-folds cross validation ($n = 3873$). At the default probability threshold of 0.5, the mean sensitivity was 0.05 ($SD = 0.02$) and the mean specificity was 0.99 ($SD = 0.005$). The mean positive predictive value is 0.45, meaning that 45% of positive predictions were correct. The mean negative predictive value is 0.89, meaning that 89% of negative predictions were correct. However, the PPV and NPV are strongly influenced by the prevalence in this cohort (approximately 12%), a higher prevalence would expected to yield a higher PPV and a lower NPV even with the same underlying model performance.

The mean area under the ROC curve across the folds was 0.73 ($SD = 0.03$), and the AUC of the ROC based on the pooled cross-validated predictions was 0.73, see Figure. This means that a randomly chosen patient with an event receives a higher predicted probability than a randomly chosen patient without an event in about 73% of cases. This is better than the random classifier ($AUC = 0.5$), but far from perfect discrimination ($AUC = 1$). Furthermore, the ROC curve shows how sensitivity and specificity trade off as the threshold varies. For example, for a sensitivity of roughly 0.8, a specificity of roughly 0.5 should be accepted.

```
# same proportion in every fold, check if that is correct
# way of validating the final model of part 1 with no external data available

library(caret)
library(pROC)

set.seed(1234)

folds <- createFolds(smart1$EVENT, k = 5, returnTrain = TRUE) # ensures by default that EVENT
print(str(folds))
```

List of 5

```
$ Fold1: int [1:3099] 1 2 3 4 5 6 7 8 9 10 ...
$ Fold2: int [1:3098] 1 2 3 4 5 6 7 8 9 10 ...
$ Fold3: int [1:3099] 2 7 9 10 11 12 13 14 16 17 ...
$ Fold4: int [1:3098] 1 3 4 5 6 8 10 12 13 14 ...
$ Fold5: int [1:3098] 1 2 3 4 5 6 7 8 9 11 ...
NULL
```

```
train_control <- trainControl(
  method = "cv",
  number = 5,
  index = folds,
  classProbs = TRUE, # needed to get predicted probabilities, not just class
  summaryFunction = twoClassSummary, # computes ROC/AUC, sensitivity, specificity
  savePredictions = "final" # keeps held-out predictions for plotting the ROC curve
)

model_glm <- train(
  EVENT ~ AGE + SEX + BMI + SYSTH + HDL + DIABETES + HISTCAR2 +
    HOMOC + log(CREAT) + STENOSIS + IMT + SMOKING + alcohol + albumin,
  data = smart1,
  method = "glm",
  family = "binomial",
  trControl = train_control,
  metric = "ROC" # tells caret to select/report based on AUC
)

# View CV performance
print(model_glm)
```

Generalized Linear Model

```
3873 samples
  14 predictor
  2 classes: 'no', 'yes'
```

```
No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 3099, 3098, 3099, 3098, 3098
Resampling results:
```

```
ROC      Sens      Spec
0.7321867 0.9912079 0.05434783
```

```
print(model_glm$results)      # mean ROC (AUC), Sens, Spec across the 5 fold (sensitivity and
```

```
parameter      ROC      Sens      Spec      ROCSD      SensSD      SpecSD
1      none 0.7321867 0.9912079 0.05434783 0.025276 0.005487882 0.02033509
```

```
# Extract the saved CV predictions
cv_preds <- model_glm$pred

# Confusion matrix at 0.5, using the labels caret stored
confusionMatrix(cv_preds$pred, cv_preds$obs, positive = "yes") # sensitivity for events
```

Confusion Matrix and Statistics

```

      Reference
Prediction  no  yes
      no  3383  435
      yes   30   25

      Accuracy : 0.8799
      95% CI : (0.8693, 0.89)
      No Information Rate : 0.8812
      P-Value [Acc > NIR] : 0.6099

      Kappa : 0.0736

      McNemar's Test P-Value : <2e-16

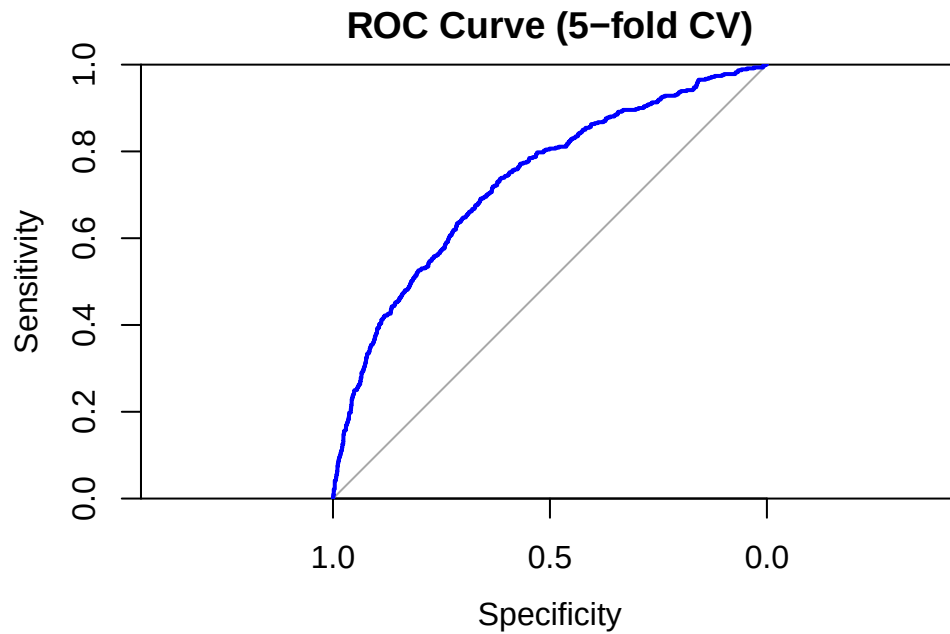
      Sensitivity : 0.054348
```

Specificity : 0.991210
Pos Pred Value : 0.454545
Neg Pred Value : 0.886066
Prevalence : 0.118771
Detection Rate : 0.006455
Detection Prevalence : 0.014201
Balanced Accuracy : 0.522779

'Positive' Class : yes

```
# Build ROC object (probability of "yes" vs actual class)
roc_obj <- roc(response = cv_preds$obs,
               predictor = cv_preds$yes,
               levels = c("no", "yes"))

# Plot
plot(roc_obj,
     main = "ROC Curve (5-fold CV)",
     col = "blue",
     lwd = 2,
     xlim = c(1, 0),
     xaxs = "i",
     yaxs = "i",
     xlab = "Specificity",
     ylab = "Sensitivity")
```



```
auc(roc_obj)
```

Area under the curve: 0.7309

Question B

```
# Calibration

# Split + fit (same model as 1c)
set.seed(2026)
smart_split <- initial_split(smart1, prop = 0.75, strata = EVENT)
train_data  <- training(smart_split)
test_data   <- testing(smart_split)

fit <- glm(EVENT ~ AGE + SEX + BMI + SYSTH + HDL + DIABETES + HISTCAR2 +
            HOMOC + log(CREAT) + STENOSIS + IMT + SMOKING + alcohol + albumin,
            data = train_data, family = binomial)

# Predicted P(EVENT = "yes") on the test set + observed class
testProbs <- tibble(
```

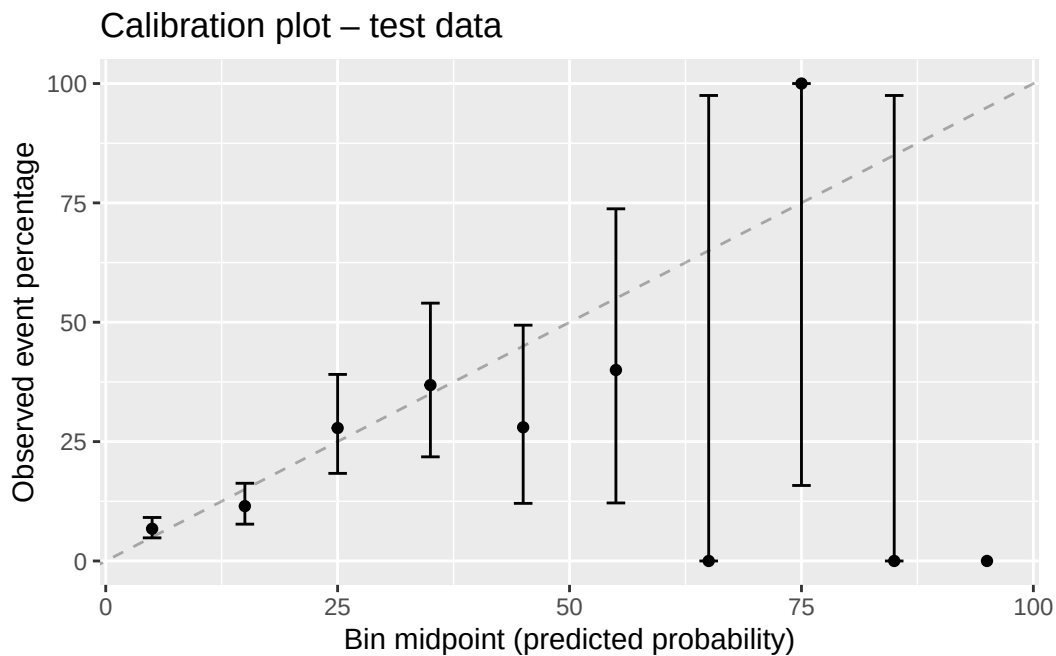
```

.pred_yes = predict(fit, newdata = test_data, type = "response"),
Class      = test_data$EVENT
)

# Calibration plot on the test data (caret, slide 31)
cal_obj <- caret::calibration(Class ~ .pred_yes, data = testProbs,
                              class = "yes", cuts = 10)

ggplot(cal_obj) +
  labs(title = "Calibration plot - test data",
       x = "Bin midpoint (predicted probability)",
       y = "Observed event percentage")

```



```

# Number of patients and events per bin
bins <- cut(testProbs$.pred_yes, breaks = seq(0, 1, 0.1), include.lowest = TRUE)
data.frame(
  patients      = as.vector(table(bins)),
  events        = as.vector(tapply(testProbs$Class == "yes", bins, sum)),
  observed_pct  = round(as.vector(tapply(testProbs$Class == "yes", bins, mean)) * 100, 1),
  row.names     = levels(bins)
)

```

```
patients events observed_pct
```

[0,0.1]	578	39	6.7
(0.1,0.2]	235	27	11.5
(0.2,0.3]	79	22	27.8
(0.3,0.4]	38	14	36.8
(0.4,0.5]	25	7	28.0
(0.5,0.6]	10	4	40.0
(0.6,0.7]	1	0	0.0
(0.7,0.8]	2	2	100.0
(0.8,0.9]	1	0	0.0
(0.9,1]	0	NA	NA

The calibration of the logistic regression model is evaluated using two calibration plots. First, the model is fitted on the training data (75%) and the predicted probabilities are calculated for the test data (25%). The test patients are divided into 10 bins based on their predicted risk, and for each bin the observed event percentage is plotted with a 95% confidence interval (see the figure above). If the model is well calibrated, the points lie on the diagonal. In the first two bins (0–10% and 10–20%), which contain most of the patients, the observed event percentages are about 7% and 11%. These are close to the diagonal with narrow confidence intervals, meaning that for low-risk patients the predicted risks agree with the observed event rates. The bins at 25% and 35% are also close to the diagonal. At 45% and 55% the observed percentages (28% and 40%) are lower than predicted, but the confidence intervals are wide and still include the diagonal. Above 60% the bins contain only one or two patients (confidence intervals from 0 to 97.5%), and the 95% bin is empty, which our caret shows as 0%. These bins give no information about the calibration of the model. However, because the test set is small, the calibration for high-risk patients cannot be judged from this plot. Therefore, we also evaluate using 5-fold cross-validation on the full data set ($n = 3873$).

```
# Cross-validated calibration with intercept & slope

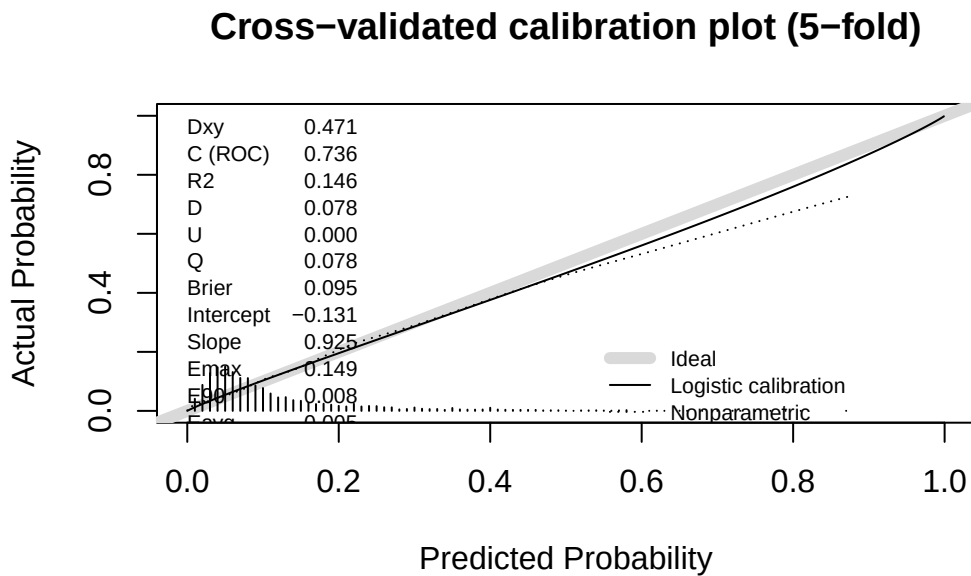
form <- EVENT ~ AGE + SEX + BMI + SYSTH + HDL + DIABETES + HISTCAR2 +
  HOMOC + log(CREAT) + STENOSIS + IMT + SMOKING + alcohol + albumin

set.seed(2026)
K      <- 5
folds  <- sample(rep(1:K, length.out = nrow(smart1)))
cvprob <- rep(NA_real_, nrow(smart1))

for (k in 1:K) {
  m <- glm(form, data = smart1[folds != k, ], family = binomial)
  cvprob[folds == k] <- predict(m, newdata = smart1[folds == k, ], type = "response")
}
```

```
# val.prob() needs the outcome as 0/1: factor "no"/"yes" -> 1/2 -> 0/1
y01 <- as.numeric(smart1$EVENT) - 1

cal_stats <- val.prob(cvprob, y01) # draws the plot and returns statistics
title("Cross-validated calibration plot (5-fold)")
```



```
# Summary of main statistics separate:
round(cal_stats[c("Intercept", "Slope", "C (ROC)", "Brier")], 3)
```

Intercept	Slope	C (ROC)	Brier
-0.131	0.925	0.736	0.095

```
# 95% confidence intervals for the calibration intercept and slope
lp <- qlogis(cvprob) # cross-validated linear predictor

recal <- glm(y01 ~ lp, family = binomial)
round(cbind(estimate = coef(recal), confint.default(recal)), 3)
```

	estimate	2.5 %	97.5 %
(Intercept)	-0.131	-0.349	0.086
lp	0.925	0.817	1.034

```
# Calibration-in-the-large: intercept with slope fixed at 1
citl <- glm(y01 ~ offset(lp), family = binomial)
round(cbind(estimate = coef(citl), confint.default(citl)), 3)
```

```
              estimate  2.5 % 97.5 %
(Intercept)         0 -0.104  0.103
```

The calibration slope is 0.925 (95% CI 0.817–1.034) and the calibration intercept is -0.131 (95% CI -0.349–0.086). For good calibration the slope should be close to 1 and the intercept close to 0. A slope below 1 means that the predictions are somewhat too extreme, which suggests mild overfitting. However, the confidence interval includes 1, so there is no clear evidence of overfitting. Because the intercept is estimated together with the slope, it is hard to interpret on its own. Therefore, the calibration-in-the-large (the intercept with the slope fixed at 1) is also estimated. This is 0.000 (95% CI -0.104–0.103), meaning that on average the predicted risks agree with the observed event rate. The Spiegelhalter test also shows no significant miscalibration ($z = 0.556$, $p = 0.578$).

The histogram at the bottom of the plot shows that almost all patients have a predicted risk below 0.2, which is in line with the prevalence of around 12%. In this range the nonparametric curve follows the ideal line. Above a predicted risk of about 0.45 the curve falls below the ideal line, with a maximum difference ($E_{\max}$) of 0.149. This means that the model overestimates the risk for the small group of high-risk patients, which is in line with the slope below 1.

In conclusion, the model is well calibrated for the large majority of patients, who have a low predicted risk. For the few patients with a high predicted risk, the model overestimates the risk, so these predictions should be interpreted with care.

Question C

Below, a comparison between the predictive performance of the logistic regression model and the Linear discriminant analysis model is made. For both analysis, the same predictors are used.

Below an Linear discriminant analysis is executed.

```
library(caret)
library(MASS)
library(pROC)

set.seed(1234)
```

```

model_lda <- train(
  EVENT ~ AGE + SEX + BMI + SYSTH + HDL + DIABETES + HISTCAR2 +
    HOMOC + log(CREAT) + STENOSIS + IMT + SMOKING + alcohol + albumin,
  data = smart1,
  method = "lda",
  trControl = train_control,
  metric = "ROC"
)

print(model_lda)

```

Linear Discriminant Analysis

```

3873 samples
  14 predictor
  2 classes: 'no', 'yes'

```

```

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 3099, 3098, 3099, 3098, 3098
Resampling results:

```

ROC	Sens	Spec
0.7325744	0.9783133	0.1195652

Both models are validated using 5-fold-cross-validation. For each fold the area under the ROC (AUC) is determined. The mean AUC of these five values is used as estimate of the performance of the prediction of the model.

```

# Mean performance across the 5 folds
model_lda$results

```

	parameter	ROC	Sens	Spec	ROCSD	SensSD	SpecSD
1	none	0.7325744	0.9783133	0.1195652	0.03375114	0.00870781	0.01537189

```

# Performance separately for each fold
model_lda$resample

```

	ROC	Sens	Spec	Resample
--	-----	------	------	----------

1	0.6773078	0.9706745	0.1413043	Fold1
2	0.7297727	0.9780381	0.1086957	Fold2
3	0.7383017	0.9692082	0.1086957	Fold3
4	0.7519734	0.9838946	0.1086957	Fold4
5	0.7655166	0.9897511	0.1304348	Fold5

```
cat("Linear discriminant analysis model: \n")
```

Linear discriminant analysis model:

```
mean_lda_auc <- model_lda$results$ROC
mean_lda_auc
```

```
[1] 0.7325744
```

```
cat("Logistic regression model: \n")
```

Logistic regression model:

```
mean_glm_auc <- model_glm$results$ROC
mean_glm_auc
```

```
[1] 0.7321867
```

the mean AUC of the performed cross validation of the LDA model is 0.733 and 0.732 for the logistic regression model. So, the difference in AUC between the LDA and the logistic regression model is very small. However, this does not take uncertainty into account. Below we will find the 95% confidence interval of the AUC for both models.

```
glm_preds_compare <- model_glm$pred %>%
  dplyr::select(
    rowIndex,
    Resample,
    obs,
    glm_prob = yes
  )

lda_preds_compare <- model_lda$pred %>%
  dplyr::select(
```

```

    rowIndex,
    Resample,
    obs,
    lda_prob = yes
  )

compare_preds <- inner_join(
  glm_preds_compare,
  lda_preds_compare,
  by = c("rowIndex", "Resample", "obs")
)

```

```

roc_glm_compare <- roc(
  response = compare_preds$obs,
  predictor = compare_preds$glm_prob,
  levels = c("no", "yes")
)

roc_lda_compare <- roc(
  response = compare_preds$obs,
  predictor = compare_preds$lda_prob,
  levels = c("no", "yes")
)
cat("Logistic regression model\n")

```

Logistic regression model

```
cat("AUC:", round(as.numeric(auc(roc_glm_compare)), 4), "\n")
```

AUC: 0.7309

```

cat(
  "95% CI:",
  round(ci.auc(roc_glm_compare)[1], 4), "-",
  round(ci.auc(roc_glm_compare)[3], 4), "\n\n"
)

```

95% CI: 0.7058 - 0.7561

```
cat("Linear discriminant analysis model\n")
```

Linear discriminant analysis model

```
cat("AUC:", round(as.numeric(auc(roc_lda_compare)), 4), "\n")
```

AUC: 0.7304

```
cat(
  "95% CI:",
  round(ci.auc(roc_lda_compare)[1], 4), "-",
  round(ci.auc(roc_lda_compare)[3], 4), "\n"
)
```

95% CI: 0.7051 - 0.7557

The 95% confidence interval of the AUC of the pooled out-of-fold validation is calculated for both models. In this method the predictions for all 5 folds were combined and one AUC is calculated. Instead of taking the average of the 5 AUCs. Which causes a small variation between this 2 methods.

Using this out of fold method. For the logistic regression we have a estimated area under the curve of 0.7309, with a 95% confidence interval of (0.7058 ; 0.7561). For the LDA method we estimate the area under the curve 0.7304, with a 95% confidence interval of (0.7051;0.7557). So, the pooled AUCs of both models are almost identical with overlapping 95% confidence intervals. So with a 95% confidence interval there is no clear evidence of a difference between the 2 models.